

Compromised intestinal epithelial barrier induces adaptive immune compensation that protects from colitis.

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Mice lacking junctional adhesion molecule A (JAM-A, encoded by F11r) exhibit enhanced intestinal epithelial permeability, bacterial translocation, and elevated colonic lymphocyte numbers, yet do not develop colitis. To investigate the contribution of adaptive immune compensation in response to increased intestinal epithelial permeability, we examined the susceptibility of F11r(-/-)Rag1(-/-) mice to acute colitis. Although negligible contributions of adaptive immunity in F11r(+/+)Rag1(-/-) mice were observed, F11r(-/-)Rag1(-/-) mice exhibited increased microflora-dependent colitis. Elimination of T cell subsets and cytokine analyses revealed a protective role for TGF- β -producing CD4(+) T cells in F11r(-/-) mice. Additionally, loss of JAM-A resulted in elevated mucosal and serum IgA that was dependent upon CD4(+) T cells and TGF- β . Absence of IgA in F11r(+/+)IgA(-/-) mice did not affect disease, whereas F11r(-/-)IgA(-/-) mice displayed markedly increased susceptibility to acute injury-induced colitis. These data establish a role for adaptive immune-mediated protection from acute colitis under conditions of intestinal epithelial barrier compromise.